

Algebra 1
Unit 10
Lesson 2 Practice

Name _____
Date _____
Period _____

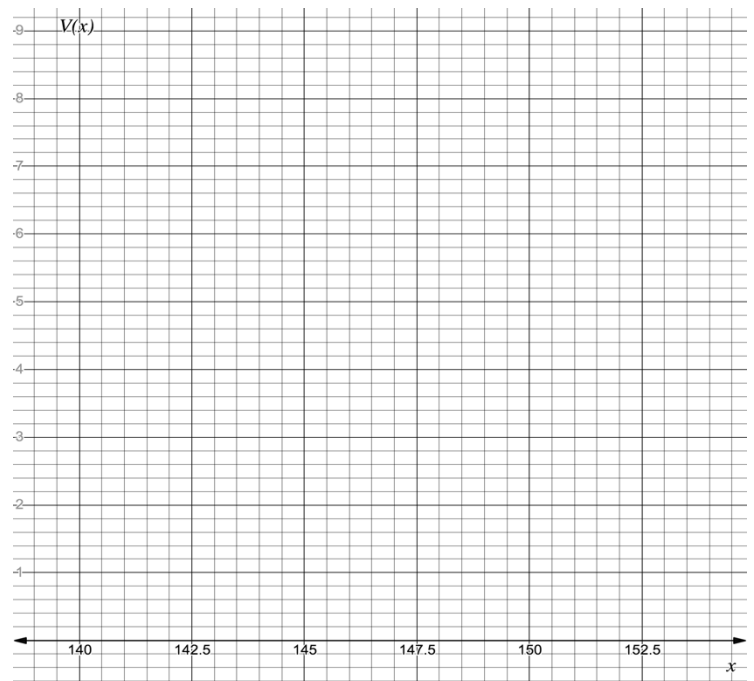
Use the following context to answer questions 1-5.

The average mass of a baseball is $145\frac{1}{2}$ grams (g) with a tolerance of $3\frac{1}{2}$ g. If x is the mass of the baseball and V is the variation from $145\frac{1}{2}$ g., the function $V(x) = \left|x - 145\frac{1}{2}\right|$ can be used to find the amount of variation.

- Determine the maximum and minimum x -values.
- Complete the table for $V(x) = \left|x - 145\frac{1}{2}\right|$. Use the minimum x -value in the first column, the maximum x -value in the last column, and add $1\frac{3}{4}$ to each x -value in between.

x					
y					

- Sketch the graph of the absolute value function.
- Interpret the vertex of $V(x) = \left|x - 145\frac{1}{2}\right|$.
- Interpret the axis of symmetry of $V(x) = \left|x - 145\frac{1}{2}\right|$.



Use the following context to answer questions 6-10.

The clearance below a bridge is 185.5 feet (ft) above sea level. According to the distribution of tidal phases, the water under the bridge can fluctuate up to 3 feet above or below the normal depth. If x is the amount of clearance the bridge has above the water, in ft, and W is the variation in water depth away from sea level, in ft, the equation to find the amount of variation is $W(x) = |x - 185.5|$.

6. Determine the maximum and minimum clearance between the bridge and water.

7. Complete the table for $W(x) = |x - 185.5|$.

x	182.5	183.5	184.5	185.5	186.5	187.5	188.5
y							

8. Sketch the graph of the absolute value function.

9. Interpret the vertex of $W(x) = |x - 185.5|$.

10. Interpret the axis of symmetry of $W(x) = |x - 185.5|$.

